UNIT - I

LAMINAR & TURBULENT FLOW IN PIPES

Subject: Hydraulics and Hydraulic Machinery (23A01404)

Syllabus at a glance

- Laminar flow through circular pipes, annulus and parallel plates
- Stokes' law; measurement of viscosity
- Reynolds experiment; transition from laminar to turbulent flow
- Resistance to flow in smooth and rough pipes; Moody's diagram
- Introduction to boundary layer theory

1. Basics: Viscosity and Types of Flow

1.1 Newton's law of viscosity

Shear stress between adjacent fluid layers is proportional to the **velocity gradient**.

$$\tau = \mu \, (du/dy)$$

- μ = dynamic viscosity (Pa.s or $N.s/m^2$; 1 poise = 0.1 Pa.s). Kinematic viscosity $\nu = \mu/\rho$ (m^2/s; 1 stoke = 10^{-4} m^2/s).
- Viscosity of **liquids decreases** with temperature; of **gases increases**.

1.2 Laminar and turbulent flow

Basis	Laminar flow	Turbulent flow
Path of particles	Smooth, parallel layers (streamlines)	Irregular, zig-zag, mixing, eddies
Shear stress	Only viscous: $\tau = \mu \, du/dy$	Viscous + turbulent (Reynolds) stress
Velocity profile	Parabolic	Flatter, fuller near the core
Reynolds number	Re < 2000 (pipes)	Re > 4000 (pipes)
Head loss	hf V	hf V^1.75 to 2
Occurs in	Viscous fluids, low velocity, small pipes	Water, air at ordinary velocities

1.3 Reynolds number

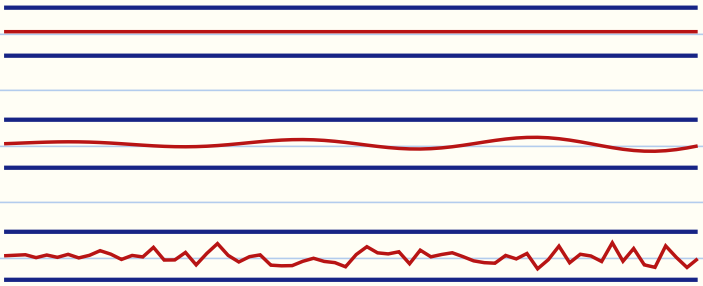
$$Re = \rho V D / \mu = V D / \nu$$

- Ratio of **inertia force** to **viscous force**. For pipes: laminar if **Re < 2000**, transition 2000 to 4000, turbulent if **Re > 4000**.

2. Reynolds Experiment

2.1 Set-up and procedure

- A large glass tank of water, a horizontal glass tube with a bell-mouthed entry and a regulating valve; a **dye container** injects a fine filament of coloured dye at the entry.



Laminar flow (Re < 2000)

Transition (2000 < Re < 4000)

Turbulent flow (Re > 4000)

- At **low velocity** the dye moves as a straight thread (laminar). As velocity is raised the thread starts to wave (transition), and at **high velocity** it breaks up and mixes completely (turbulent).

2.2 Observations

- **Lower critical velocity**: velocity (Re about 2000) below which flow is always laminar. **Upper critical velocity**: Re about 4000 (up to 10,000 in very quiet conditions) above which flow is turbulent.
- Transition: flow is unstable; small disturbances make it turbulent. Depends on **roughness, entry conditions and disturbances**.

2.3 Critical velocity

$$V_c = Re_c \cdot \nu / D \quad \text{(for pipe)}$$

Example: Water $\nu = 10^{-6}$ m²/s flows in D = 25 mm pipe. Lower critical velocity = 2000 x 10^{-6} / 0.025 = **0.08 m/s**.

3. Laminar Flow Through a Circular Pipe (Hagen-Poiseuille)

3.1 Assumptions

- Steady, incompressible, fully developed laminar flow; pipe horizontal and long; no slip at wall.

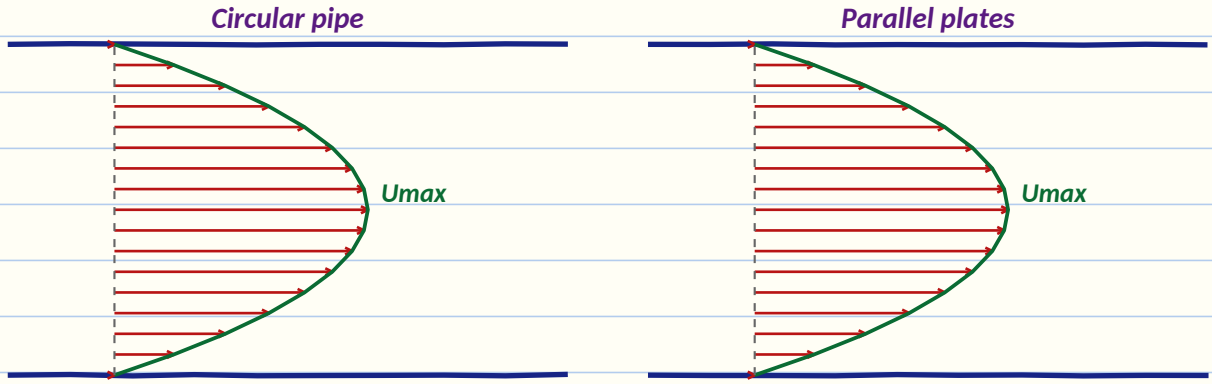
3.2 Derivation (summary)

- Take a coaxial fluid cylinder of radius r and length dx. **Pressure force = shear force** :

$$\tau = -(\partial p/\partial x)(r/2) \quad \text{(shear stress varies linearly: zero at axis, maximum at wall)}$$

- Using $\tau = -\mu\, du/dr$ and integrating with $u = 0$ at $r = R$ (no slip):

$$u = (1/4\mu)(-\partial p/\partial x)(R^2 - r^2) \quad \text{(parabolic profile)}$$



Pipe: Umax = 2 x average velocity

Plates: Umax = 1.5 x average velocity

3.3 Important results

$$u_{max}\ (\text{at } r = 0) = (1/4\mu)(-\partial p/\partial x)\, R^2 = 2 \times V_{avg}$$

$$V_{avg} = (1/8\mu)(-\partial p/\partial x)\, R^2 = u_{max}/2$$

$$Q = \pi R^4 (-\partial p/\partial x)/8\mu = \pi D^4 \Delta p/(128\,\mu L) \quad \text{(Hagen-Poiseuille)}$$

$$\Delta p = 32\,\mu V L/D^2 \;;\; h_f = 32\,\mu V L/(\rho g D^2) = 128\,\mu Q L/(\pi \rho g D^4)$$

$$\text{Wall shear stress } \tau_0 = 8\,\mu V/D = (\Delta p/L)(D/4)$$

$$\text{Darcy friction factor (laminar): } f = 64/Re$$

- Radius at which local velocity = average velocity: **r = 0.707 R** .

3.4 Solved Problem 1

Oil ($\mu = 0.1$ Pa.s, $\rho = 900$ kg/m^3) flows at 1 L/s through a 50 mm pipe, L = 100 m. Find Re, pressure drop, head loss, u_{max} and wall shear stress.

- $A = \pi/4 \times 0.05^2 = 1.9635 \times 10^{-3}$ m^2; $V = 0.001 / 1.9635 \times 10^{-3} =$ **0.509 m/s** .

$$Re = 900 \times 0.509 \times 0.05 / 0.1 = 229 \quad (< 2000, \text{ laminar})$$

$$\Delta p = 32 \times 0.1 \times 0.509 \times 100 / 0.05^2 = 65{,}190 \text{ Pa} \approx 65.2 \text{ kPa}$$

$$h_f = 65{,}190 / (900 \times 9.81) = 7.38 \text{ m of oil}$$

- $u_{max} = 2V =$ **1.02 m/s** ; $\tau_0 = 8 \times 0.1 \times 0.509 / 0.05 =$ **8.15 Pa** .
- Check with f: $h_f = f L V^2 / 2gD = (64/229) \times 100 \times 0.259 / (2 \times 9.81 \times 0.05) = 7.38$ m.

4. Laminar Flow Between Parallel Plates

4.1 Both plates fixed (plane Poiseuille flow)

- Plates separated by B; y measured from the lower plate.

$$u = (1/2\mu)(-\partial p/\partial x)(B y - y^2)$$

$$u_{max} \text{ (at } y = B/2) = (-\partial p/\partial x) B^2 / 8\mu \; ; \; V_{avg} = (2/3) u_{max}$$

$$\text{Discharge per unit width: } q = (-\partial p/\partial x) B^3 / 12\mu$$

$$\Delta p = 12 \mu V L / B^2 \; ; \; h_f = 12 \mu V L / (\rho g B^2)$$

$$\text{Shear stress at plates: } \tau_0 = (-\partial p/\partial x) B / 2 = 6 \mu V / B$$

4.2 One plate moving (Couette flow)

- Upper plate moves with U, lower fixed. With zero pressure gradient: **$u = U y / B$** (linear profile), $\tau = \mu U / B$.
- With pressure gradient (Couette + Poiseuille): **$u = U y/B + (1/2\mu)(-\partial p/\partial x)(By - y^2)$** . Flow can reverse near the fixed plate if the pressure gradient is adverse.

4.3 Solved Problem 2

Oil (μ = 0.08 Pa.s) flows between two fixed plates 10 mm apart with pressure gradient $-\partial p/\partial x$ = 2000 Pa/m. Find u_{max}, V_{avg}, discharge per metre width and wall shear stress.

$$u_{max} = 2000 \times (0.01)^2 / (8 \times 0.08) = 0.3125 \text{ m/s}$$

$$V_{avg} = (2/3) \times 0.3125 = 0.208 \text{ m/s}; \quad q = 2000 \times (0.01)^3 / (12 \times 0.08) = 0.00208 \text{ m}^3\text{/s per m}$$

$$\tau_0 = 2000 \times 0.01 / 2 = 10 \text{ Pa}$$

5. Laminar Flow Through an Annulus

5.1 Set-up

- Concentric annulus: **outer radius R** , **inner radius r0** ; flow along axis under pressure gradient. No-slip on both walls ($u = 0$ at $r = R$ and at $r = r_0$).

5.2 Results

$$u = (1/4\mu)(-\partial p/\partial x)\left[R^2 - r^2 + (R^2 - r_0^2) \ln(r/R) / \ln(R/r_0) \right]$$

$$Q = (\pi / 8\mu)(-\partial p/\partial x)\,[\,R^4 - r0^4 - (R^2 - r0^2)^2 / \ln(R/r0)\,]$$

$$\text{Mean velocity } V = Q / [\pi (R^2 - r0^2)]$$

- Maximum velocity occurs at $r = \sqrt{[(R^2 - r0^2) / (2 \ln(R/r0))]}$; zero shear at this radius.
- Applications: journal bearings, oil in annular gap of pistons, flow in concentric tube heat exchangers.

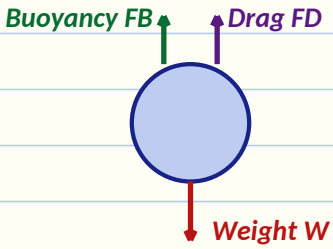
5.3 Hydraulic radius view

- For an annulus: hydraulic diameter $D_h = 4A / P = 2(R - r0)$; the limiting case $r0 \to 0$ gives the circular pipe results.

6. Stokes' Law and Falling-Sphere Viscometer

6.1 Stokes' law

For a small sphere moving slowly (Re < 0.2) through a viscous fluid, the drag force is $F_D = 3\pi\mu d V$.



At terminal velocity: W = FB + FD

Stokes drag: FD = 3 pi mu d V

Valid for Re < 0.2 (creeping flow)

$$\text{Terminal velocity: } V = g\,d^2\,(\rho s - \rho) / (18\,\mu)$$

- At terminal velocity: weight W = buoyancy FB + drag FD. Drag coefficient $C_D = 24 / Re$ for Re < 0.2.

6.2 Solved Problem 3

A steel ball (d = 2 mm, ρs = 7800) falls in glycerine (μ = 0.8 Pa.s, ρ = 1260). Find terminal velocity.

$$V = 9.81 \times (0.002)^2 \times (7800 - 1260) / (18 \times 0.8) = 0.0178 \text{ m/s}$$

- Check: $Re = 1260 \times 0.0178 \times 0.002 / 0.8 = 0.056$ (< 0.2), so Stokes' law is valid.

7. Measurement of Viscosity

7.1 Methods

Instrument	Principle	Formula
Capillary tube viscometer	Hagen-Poiseuille flow in a tube	$\mu = \pi d^4 \Delta p / (128 Q L)$
Falling sphere viscometer	Stokes' law	$\mu = g d^2 (\rho s - \rho) / (18 V)$
Rotating cylinder (Couette)	Torque on coaxial cylinders	$\mu = T \delta / (2\pi R^3 L \omega)$
Redwood / Saybolt / Engler	Time of efflux of fixed oil volume	$\nu = A t - B / t$

- **Rotating cylinder** : fluid in thin gap δ between a fixed outer and a rotating inner cylinder; measure torque T at angular speed ω.
- **Efflux viscometers** give kinematic viscosity in empirical units (Redwood seconds, Saybolt seconds); convert with the given formula.

8. Turbulent Flow in Pipes

8.1 Nature

- Velocity at a point fluctuates: $u = \bar{u} + u'$. **Reynolds stress** $= -\rho \overline{u'v'}$ (extra shear due to eddy mixing).
- Prandtl's mixing-length theory: $\tau_t = \rho l^2 (du/dy)^2$, with $l = \kappa y$ ($\kappa = 0.4$, von Karman constant).

8.2 Regions near the wall

- **Viscous (laminar) sub-layer** : very thin, thickness $\delta' = 11.6 \nu / v$; buffer zone ; **and** turbulent core .
- Shear (friction) velocity: $v = \sqrt{\tau_0 / \rho} = V \sqrt{f/8}$.

8.3 Smooth and rough pipes

Condition	Criterion (k = roughness height)
Hydraulically smooth	$k / \delta' < 0.25$ (roughness is submerged in the sub-layer)
Transition	$0.25 < k / \delta' < 6$
Hydraulically rough	$k / \delta' > 6$ (roughness projects out, causes eddies)

8.4 Velocity distribution laws

Smooth: $u / v^* = 5.75 \log_{10}(y v^* / \nu) + 5.5$

Rough: $u / v^* = 5.75 \log_{10}(y / k) + 8.5$

Velocity defect law: $(u_{max} - u) / v^* = 5.75 \log_{10}(R / y)$

- Power-law (approx): **$u / u_{max} = (y/R)^{1/n}$**, $n = 7$ for Re about 10^5 (1/7th power law); mean velocity $V \approx 0.817\, u_{max}$ for $n = 7$.

9. Resistance to Flow: Friction Factor

9.1 Darcy-Weisbach equation

$h_f = f L V^2 / (2 g D)$; $f = 8 \tau_0 / (\rho V^2)$

- (Fanning friction factor $f' = f / 4$ in some books.) Also $h_f = 4 f' L V^2 / (2gD)$.

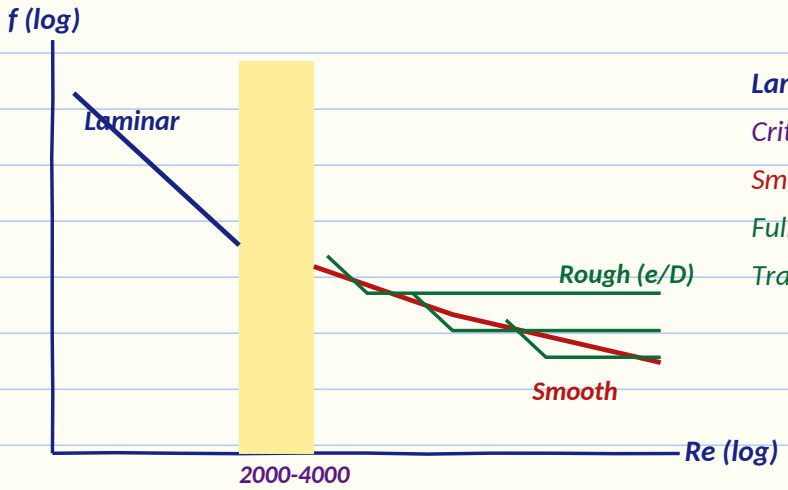
9.2 Friction factor correlations

Flow regime	Formula
Laminar (Re < 2000)	$f = 64 / Re$
Smooth turbulent, 4000 < Re < 10^5	Blasius: $f = 0.316 / Re^{0.25}$
Smooth turbulent, any Re	Prandtl: $1/\sqrt{f} = 2 \log_{10}(Re \sqrt{f}) - 0.8$
Fully rough	von Karman: $1/\sqrt{f} = 2 \log_{10}(3.7 D / k)$
Transition (smooth and rough)	Colebrook-White: $1/\sqrt{f} = -2 \log_{10}[k/(3.7D) + 2.51/(Re \sqrt{f})]$

- Explicit approximation: **Swamee-Jain**: $f = 0.25 / [\log_{10}(k/(3.7D) + 5.74/Re^{0.9})]^2$.

9.3 Moody's diagram

- A plot of **f (y-axis) versus Re (x-axis, log scale)** for various relative roughness k/D.



Laminar: f = 64 / Re

Critical zone: Re 2000 to 4000

Smooth (Blasius): f = 0.316 / Re^0.25

Fully rough: f depends on e/D only

Transition: f depends on Re and e/D

- **Laminar zone** : straight line f = 64/Re, independent of roughness. **Critical zone** (2000 to 4000): uncertain. **Smooth pipe curve** ; **transition zone** : curves of constant k/D drop and flatten; **fully rough zone** : horizontal lines, f depends only on k/D (hf V^2).
- **Usage** : compute Re and k/D, read f, then find $h_f = f\,L\,V^2 / 2gD$.

9.4 Solved Problems

Problem 4: Water ($\nu = 10^{-6}$) flows at 0.5 L/s in a 25 mm pipe. Find Re and the nature of flow.

- $V = 0.0005 / 4.909 \times 10^{-4} = 1.019$ m/s; **$Re = 1.019 \times 0.025 / 10^{-6} = 25{,}460$** (turbulent).

Problem 5: D = 0.1 m, L = 200 m, V = 2 m/s, f = 0.02. Find head loss.

$h_f = 0.02 \times 200 \times 2^2 / (2 \times 9.81 \times 0.1) = 16 / 1.962 = 8.15$ m

Problem 6: Smooth pipe at Re = 50,000. Find f (Blasius).

f = 0.316 / (50,000)^0.25 = 0.316 / 14.95 = 0.0211

Problem 7: Power to overcome friction. Q = 0.02 m^3/s water, hf = 8.15 m.

$P = \rho g Q h_f = 1000 \times 9.81 \times 0.02 \times 8.15 = 1599$ W ≈ 1.6 kW

9.5 Minor losses (other resistances in pipes)

Loss	Formula
Sudden enlargement	$h_L = (V1 - V2)^2 / 2g$
Sudden contraction	$h_L = 0.5\,V2^2 / 2g$ (approximately; Cc = 0.62)
Pipe entrance	$h_L = 0.5\,V^2 / 2g$ (sharp), 0.04 to 0.1 for rounded

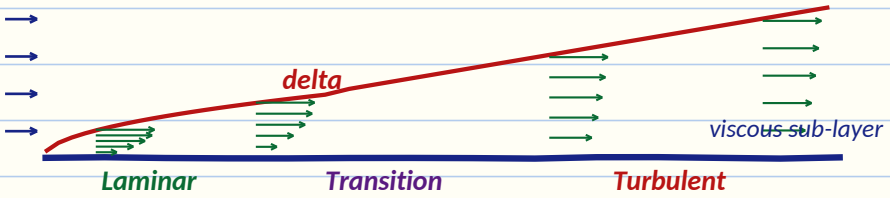
Pipe exit	$h_L = V^2 / 2g$
Bends, valves, fittings	$h_L = K V^2 / 2g$

- **Major loss** = friction along length (Darcy-Weisbach). **Minor loss** = due to change in section or direction (local disturbances). Total head loss = major + minor.

10. Introduction to Boundary Layer Theory

10.1 Concept (Prandtl, 1904)

- When a real fluid flows past a solid surface, velocity is **zero at the wall** (no-slip) and rises to the free-stream value U over a thin layer: the **boundary layer**. Outside it, flow is practically ideal (inviscid).



Boundary layer thickness delta: where u = 0.99 U

Re_x(crit) = 5 x 10^5

10.2 Thickness definitions

Term	Meaning	Formula
Boundary layer thickness δ	Distance where $u = 0.99\ U$	-
Displacement thickness δ^*	Distance wall is displaced due to velocity deficit	$\delta^* = \int (1 - u/U)\, dy$
Momentum thickness θ	Loss of momentum flux	$\theta = \int (u/U)(1 - u/U)\, dy$
Energy thickness δ^{**}	Loss of kinetic energy flux	$\int (u/U)(1 - u^2/U^2)\, dy$

10.3 Flat plate results

Quantity	Laminar ($Re_x < 5 \times 10^5$)	Turbulent
δ	$5x / \sqrt{Re_x}$ (Blasius)	$0.37x / Re_x^{0.2}$

Local Cf	0.664 / √Re_x	0.059 / Re_x^0.2
Average CD	1.328 / √Re_L	0.074 / Re_L^0.2
δ and θ	δ = 1.72 x/√Re_x; θ = 0.664 x/√Re_x	δ = δ/8 ; θ = 7δ/72

Momentum integral equation: $\tau_0 = \rho U^2 \frac{d\theta}{dx}$ (von Karman)

- Drag: $F_D = C_D \times (1/2 \rho U^2) \times A$. Transition on flat plate at $Re_x \approx 5 \times 10^5$.

10.4 Boundary layer separation

- In an **adverse pressure gradient** ($dp/dx > 0$, e.g. diverging section, rear of a cylinder), fluid near the wall loses momentum, velocity gradient at wall becomes zero and the flow **separates**, forming a wake and **form drag**.
- **Remedies**: streamlined shapes, suction or blowing, slots, rotating cylinder, turbulence promoters (dimples on golf ball).

10.5 Solved Problem 8

Air ($\nu = 1.5 \times 10^{-5}$, $\rho = 1.2$) flows at $U = 10$ m/s over a flat plate. At $x = 0.5$ m find Re_x, δ, local Cf, wall shear stress, δ and θ.

$Re_x = 10 \times 0.5 / 1.5 \times 10^{-5} = 3.33 \times 10^5$ (laminar, < 5×10^5)

- $\delta = 5 \times 0.5 / \sqrt{(3.33 \times 10^5)} = 2.5 / 577 =$ **4.33 mm** ; $C_f = 0.664 / 577 = 0.00115$.
- $\tau_0 = C_f \times 1/2 \rho U^2 = 0.00115 \times 0.5 \times 1.2 \times 100 =$ **0.069 Pa**.
- $\delta =$ **1.72 x 0.5 / 577 = 1.49 mm**; $\theta =$ **0.664 x 0.5 / 577 = 0.575 mm.**
- Drag on one side (plate 0.5 m x 1 m): $C_D = 1.328/577 = 0.0023$; $F_D = 0.0023 \times 60 \times 0.5 =$ **0.069 N**.

11. More Solved Problems

Problem 9: Laminar flow in an annulus

Oil ($\mu = 0.1$ Pa.s, $\rho = 900$) flows in an annulus with $R = 25$ mm, $r_0 = 10$ mm, under $-\partial p/\partial x = 1000$ Pa/m. Find Q, V and the radius of maximum velocity.

- $R^4 = 3.906 \times 10^{-7}$; $r_0^4 = 10^{-8}$; $(R^2 - r_0^2)^2 = (5.25 \times 10^{-4})^2 = 2.756 \times 10^{-7}$; $\ln(R/r_0) = \ln 2.5 = 0.916$.
- Bracket $= 3.906 \times 10^{-7} - 0.1 \times 10^{-7} - 3.008 \times 10^{-7} = 7.98 \times 10^{-8}$.

$Q = (\pi / 8 \times 0.1) \times 1000 \times 7.98 \times 10^{-8} = 3.13 \times 10^{-4}$ m³/s (0.313 L/s)

- Area = π (6.25 - 1) x 10^{-4} = 1.649 x 10^{-3} m^2; V = 0.19 m/s. Radius of u_max = sqrt(5.25 x 10^{-4} / 1.832) = **16.9 mm** .
- Re = 900 x 0.19 x 0.03 / 0.1 = 51, laminar (hydraulic diameter = 2(R - r0) = 30 mm).

Problem 10: Capillary tube viscometer

Tube d = 1 mm, L = 0.2 m, pressure drop 2000 Pa; 0.147 cm^3 of oil collected in 60 s. Find μ.

- Q = 0.147 x 10^{-6} / 60 = 2.45 x 10^{-9} m^3/s.

$\mu = \pi d^4 \Delta p / (128 Q L) = \pi \times 10^{-12} \times 2000 / (128 \times 2.45 \times 10^{-9} \times 0.2) = 0.1$ Pa.s

Problem 11: Rotating cylinder viscometer

R = 50 mm, L = 100 mm, gap δ = 0.5 mm, N = 60 rpm, torque T = 0.05 N.m. Find μ.

- ω = 2π x 60 / 60 = 6.283 rad/s.

$\mu = T \delta / (2\pi R^3 L \omega) = 0.05 \times 0.0005 / (2\pi \times 1.25 \times 10^{-4} \times 0.1 \times 6.283) = 0.0507$ Pa.s

Problem 12: Turbulent pipe: friction factor and sub-layer

Water (ν = 10^{-6}) in D = 0.3 m pipe, L = 500 m, k/D = 0.0005, Re = 10^6.

- Swamee-Jain: log10(1.35 x 10^{-4} + 5.74 / 10^5.4) = log10(1.58 x 10^{-4}) = - 3.80; f = 0.25 / 14.45 = **0.0173** (Moody also gives about 0.017).
- V = Re ν / D = 10^6 x 10^{-6} / 0.3 = 3.33 m/s. hf = 0.0173 x 500 x 3.33^2 / (2 x 9.81 x 0.3) = **16.3 m** .
- τ0 = ρ g hf D / 4L = 1000 x 9.81 x 16.3 x 0.3 / 2000 = 24.0 Pa; **v = sqrt(24/1000) = 0.155 m/s (check V sqrt(f/8) = 0.155).**
- δ' = 11.6 ν / v **= 11.6 x 10^{-6} / 0.155 = 0.0748 mm; k = 0.15 mm, so k/δ' = 2.0:** transition zone **(neither smooth nor fully rough).**

Problem 13: Boundary layer shape (assumed profiles)

Linear profile u/U = y/δ: **δ = ∫(1 - y/δ)dy = δ/2; θ = ∫(y/δ)(1 - y/δ)dy = δ/6; shape factor H = δ/θ = 3.**

Parabolic profile u/U = 2η - η^2 (η = y/δ): **δ = δ/3; θ = 2δ/15; H = 2.5.**

Problem 14: Turbulent boundary layer drag

Air at U = 15 m/s (ν = 1.5 x 10^{-5}) over a plate 1 m long, 1 m wide, turbulent from the leading edge.

- $Re_L = 15 \times 1 / 1.5 \times 10^{-5} = 10^6$; $Re^{0.2} = 15.85$. δ at $x = 1$ m = $0.37 / 15.85 =$ **23.3 mm** .

$CD = 0.074 / 15.85 = 0.00467$; $FD = CD \times (1/2\ \rho U^2) \times A = 0.00467 \times (0.5 \times 1.2 \times 225) \times 1 = 0.63$ N

Problem 15: Equivalent velocity profile check (1/7th power law)

Pipe R = 0.1 m, $u_{max} = 3$ m/s, $n = 7$. Velocity at $y = 0.02$ m from wall: $u = 3 \times (0.02/0.1)^{(1/7)} = 3 \times 0.795 =$ **2.39 m/s** ; mean $V = 0.817 \times 3 = 2.45$ m/s.

12. Quick Revision and Important Questions

Laminar: $\tau = \mu\ du/dy$; $f = 64/Re$; umax = 2V (pipe), 1.5V (plates). Annulus: no slip on both walls. Stokes: $F = 3\pi\mu dV$ ($Re < 0.2$). $Re < 2000$ laminar, > 4000 turbulent. Blasius $f = 0.316/Re^{0.25}$. Boundary layer: $Re_x(crit) = 5 \times 10^5$.

- Derive the velocity distribution for laminar flow through a circular pipe and obtain the Hagen-Poiseuille equation. (10 M)
- Derive expressions for laminar flow between two fixed parallel plates. (10 M)
- Explain Reynolds experiment and the transition from laminar to turbulent flow. (5 M)
- State Stokes' law; explain the falling sphere and capillary viscometers. (5 M)
- Explain Moody's diagram and the variation of friction factor for smooth and rough pipes. (10 M)
- Define displacement, momentum and energy thickness. Explain boundary layer separation. (10 M)